PRE-AMC 8 MOCK TEST

Worked solutions · Choose a question to review.

Question 1 · Answer B

> **Answer (B):** Brianna is half as old as Aunt Anna, so Brianna is 21 years old. Caitlin is 5 years younger than Brianna, so Caitlin is 16 years old.

Question 2 · Answer A

Answer (A): The number 0 has no reciprocal, and 1 and -1 are their own reciprocals. This leaves only 2 and -2. The reciprocal of 2 is $\frac{1}{2}$, but 2 is not less than $\frac{1}{2}$. The reciprocal of -2 is $-\frac{1}{2}$, and -2 is less than $-\frac{1}{2}$.

Question 3 · Answer D

Answer (D): The smallest whole number in the interval is 2 because $\frac{5}{3}$ is more than 1 but less than 2. The largest whole number in the interval is 6 because 2π is more than 6 but less than 7. There are five whole numbers in the interval. They are 2, 3, 4, 5, and 6.



Question 4 · Answer C

Answer (C): If the first year of the 8-year period was the final year of a principal's term, then in the next six years two more principals would serve, and the last year of the period would be the first year of the fourth principal's term. Therefore, the maximum number of principals who can serve during an 8-year period is 4.



Question 5 · Answer A

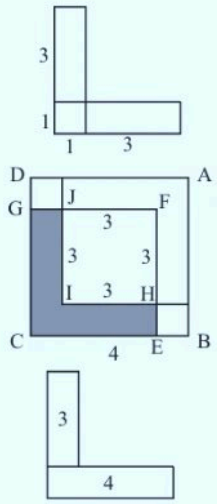
Answer (A): The L-shaped region is made up of two rectangles with area $3 \times 1 = 3$ plus the corner square with area $1 \times 1 = 1$, so the area of the L-shaped figure is $2 \times 3 + 1 = 7$.

OR

Square $FECG-$ square $FHIJ = 4 \times 4 - 3 \times 3 = 16 - 9 = 7$.

OR

The L-shaped region can be decomposed into a 4×1 rectangle and a 3×1 rectangle. So the total area is 7.



Question 6 · Answer B

Answer (B): The only way to get a negative product using three numbers is to multiply one negative number and two positives or three negatives. Only two reasonable choices exist: $(-8) \times (-6) \times (-4) = (-8) \times (24) = -192$ and $(-8) \times 5 \times 7 = (-8) \times 35 = -280$. The latter is smaller.

Question 7 · Answer D

Answer (D): The numbers on one die total $1+2+3+4+5+6=21$, so the numbers on the three dice total 63. Numbers 1, 1, 2, 3, 4, 5, 6 are visible, and these total 22. This leaves $63-22=41$ not seen.

Question 8 · Answer D

(D) At 2 seconds per dimple, it takes $300 \times 2 = 600$ seconds to paint them. Since there are 60 seconds in a minute, he will need $600 \div 60 = 10$ minutes.

Question 9 · Answer D

(D) Since their sum is to be 11, only positive factors need to be considered. Number pairs whose product is 24 are $(1, 24)$, $(2, 12)$, $(3, 8)$ and $(4, 6)$. The sum of the third pair is 11, so the numbers are 3 and 8. The larger one is 8.

Question 10 · Answer E

(E) Anjou has one-third as much money as Granny Smith, so Anjou has \$21. Elberta has \$2 more than Anjou, and $\$21 + \$2 = \$23$.

Question 11 · Answer E

(E) To make the number as small as possible, the smaller digits are placed in the higher-value positions. To make the number even, the larger even digit 4 must be the units digit. The smallest possible even number is 12394 and 9 is in the tens place.

Question 12 · Answer C

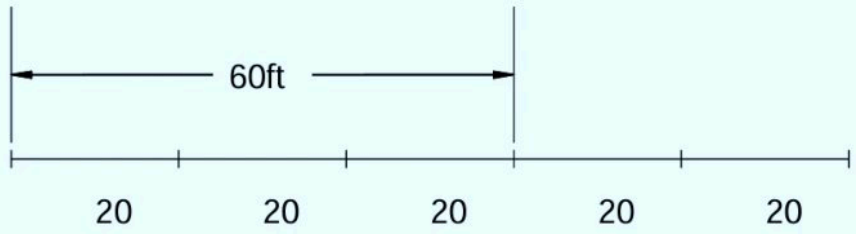
(C) Use the formula $d = rt$ (distance equals rate times time): 1088 feet per second $\times 10$ seconds $= 10880$ feet, which is just 320 feet more than two miles. Therefore, Snoopy is just about two miles from the flash of lightning.

OR

Since this is an estimate, round the speed of sound down to 1000 feet per second and the length of a mile down to 5000 feet. Then $5000 \div 1000 = 5$ seconds per mile, so in 10 seconds the sound will travel about 2 miles.

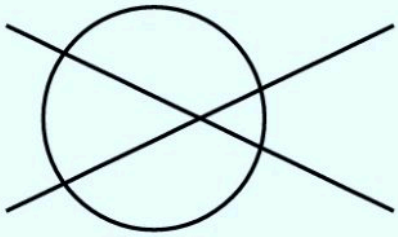
Question 13 · Answer B

(B) There are three spaces between the first tree and the fourth tree, so the distance between adjacent trees is 20 feet. There are 6 trees with five of these 20-foot spaces, so the distance between the first and last trees is 100 feet.



Question 14 · Answer D

(D) Two distinct lines can intersect in one point whereas a line can intersect a circle in two points. The maximum number 5 can be achieved if the lines and circle are arranged as shown. Note that the lines could also meet outside the circle for the same result. (Other arrangements of the lines and circle can produce 0, 1, 2, 3, or 4 points of intersection.)



Question 15 · Answer D

(D) The sum of all five numbers is $5 \times 54 = 270$. The sum of the first two numbers is $2 \times 48 = 96$, so the sum of the last three numbers is $270 - 96 = 174$. The average of the last three numbers is $\frac{174}{3} = 58$.

Question 16 · Answer C

(C) The smallest average will occur when the numbers are as small as possible. The four smallest distinct positive even integers are 2, 4, 6, and 8 and their average is 5.

Note: These numbers form an arithmetic sequence. The average of the numbers in any arithmetic sequence is the average of the first and last terms.

Question 17 · Answer A

(A) Since the total \$17 is odd, there must be an odd number of \$5 bills. One \$5 bill plus six \$2 bills is a solution, as is three \$5 bills plus one \$2 bill. Five \$5 bills exceeds \$17, so these are the only two combinations that work.

Question 18 · Answer B

> **(B)** The next palindrome is 2112. The product of its digits is $2 \cdot 1 \cdot 1 \cdot 2 = 4$.

Question 19 · Answer C

> **(C)** Since 706 days is 700 plus 6 days, it is 100 weeks plus 6 days. Friday is 6 days after Saturday.

Question 20 · Answer C

(C) The smallest prime is 2, which is a factor of every even number. Because 58 is the only even number, it has the smallest prime factor.

Question 21 · Answer B

(B) If 20% of the number is 12, the number must be 60. Then 30% of 60 is $0.30 \times 60 = 18$.

OR

Since 20% of the number is 12, it follows that 10% of the number is 6. So 30% of the number is 18.

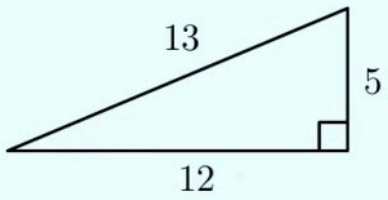
Question 22 · Answer B

(B)

$$A = \frac{1}{2}(\sqrt{144})(\sqrt{25})$$

$$A = \frac{1}{2} \cdot 12 \cdot 5$$

$$A = 30 \text{ square units}$$



Question 23 · Answer B

(B) A cube has four red faces if it is attached to exactly two other cubes. The four top cubes are each attached to only one other cube, so they have five red faces. The four bottom corner cubes are each attached to three others, so they have three red faces. The remaining six each have four red faces.

Question 24 · Answer D

(D) As given, $T = 7$. This implies that $F = 1$ and that O equals either 4 or 5. Since O is even, $O = 4$. Therefore, $R = 8$. Replacing letters with numerals gives

$$\begin{array}{cccc} & 7 & W & 4 \\ + & 7 & W & 4 \\ \hline 1 & 4 & U & 8 \end{array}$$

$W + W$ must be less than 10; otherwise, a 1 would be carried to the next column, and O would be 5. So $W < 5$. $W \neq 0$ because $W \neq U$, $W \neq 1$ because $F = 1$, $W \neq 2$ because if $W = 2$ then $U = 4 = O$, and $W \neq 4$ because $O = 4$. So $W = 3$.

The addition problem is

$$\begin{array}{cccc} & 7 & 3 & 4 \\ + & 7 & 3 & 4 \\ \hline 1 & 4 & 6 & 8 \end{array}$$

Question 25 · Answer E

(E) Because Jim has brown eyes and blond hair, none of his siblings can have both blue eyes and black hair. Therefore, neither Benjamin nor Tevyn can be Jim's sibling. Consequently, there are only three possible pairs for Jim's siblings – Nadeen and Austin, Nadeen and Sue, or Austin and Sue. Since Nadeen has different hair color and eye color from both Austin and Sue, neither can be Nadeen's sibling. So Austin and Sue are Jim's siblings. Benjamin, Nadeen and Tevyn are siblings in the other family.